d76 - Google Play Store App Analytics (start)

January 28, 2026

1 Introduction

In this notebook, we will do a comprehensive analysis of the Android app market by comparing thousands of apps in the Google Play store.

2 About the Dataset of Google Play Store Apps & Reviews

Data Source: App and review data was scraped from the Google Play Store by Lavanya Gupta in 2018. Original files listed here.

3 Import Statements

[1]:
```
import pandas as pd
```

4 Notebook Presentation

[2]:
```
# Show numeric output in decimal format e.g., 2.15
pd.options.display.float_format = '{:,.2f}'.format
```

5 Read the Dataset

[3]:
```
df_apps = pd.read_csv('apps.csv')
```

6 Data Cleaning

Challenge: How many rows and columns does `df_apps` have? What are the column names? Look at a random sample of 5 different rows with .sample().

[4]:
```
#df_apps
# df_apps.tail()
# df_apps.describe()
# df_apps.shape
df_apps.columns
#  df_apps.isna().values.any():
# df_apps.sample(5)
```

[4]:
```
Index(['App', 'Category', 'Rating', 'Reviews', 'Size_MBs', 'Installs', 'Type',
       'Price', 'Content_Rating', 'Genres', 'Last_Updated', 'Android_Ver'],
      dtype='object')
```

[5]:
```
df_apps.sample(5)
```

[5]:

	App	Category	Rating	Reviews	Size_MBs	Installs	Type	Price	Content_Rating	Genres	Last_Updated	Android_Ver
2993	Tinnitus Alleviator	MEDICAL	3.10	82	42.00	5,000	Free	0	Everyone	Medical	December 23, 2017	4.1 and up
4574	Dr. Panda & Toto's Treehouse	FAMILY	4.40	3396	9.50	50,000	Paid	$3.99	Everyone	Casual;Pretend Play	December 9, 2014	4.0 and up
5127	EL NORTE	NEWS_AND_MAGAZINES	3.40	2436	6.30	100,000	Free	0	Everyone	News & Magazines	June 26, 2018	Varies with device
4829	DataCamp - Learn R, Python & SQL	FAMILY	4.40	944	12.00	100,000	Free	0	Everyone	Education	August 1, 2018	4.1 and up
10468	Photo Editor Collage Maker Pro	PHOTOGRAPHY	4.50	1519671	6.90	100,000,000	Free	0	Everyone	Photography	February 1, 2018	Varies with device

[6]:
```
df_apps.isna().sum()
```

[6]:
```
App                0
Category           0
Rating          1474
Reviews            0
Size_MBs           0
Installs           0
Type               1
Price              0
Content_Rating     0
Genres             0
Last_Updated       0
Android_Ver        2
dtype: int64
```

[7]:
```
df_apps[df_apps.Rating.isna()]
```

[7]:

	App	Category
0	Ak Parti Yardım Toplama	SOCIAL
1	Ain Arabic Kids Alif Ba ta	FAMILY

```
2     Popsicle Launcher for Android P 9.0 launcher     PERSONALIZATION
3                          Command & Conquer: Rivals              FAMILY
4                                        CX Network            BUSINESS
…                                                …                   …
5840                                 Em Fuga Brasil              FAMILY
5862                     Voice Tables - no internet           PARENTING
6141                                 Young Speeches  LIBRARIES_AND_DEMO
7035                                 SD card backup               TOOLS
7175                      Android TV Remote Service               TOOLS

      Rating  Reviews  Size_MBs   Installs  Type   Price Content_Rating  \
0        NaN        0      8.70          0  Paid  $13.99           Teen
1        NaN        0     33.00          0  Paid   $2.99       Everyone
2        NaN        0      5.50          0  Paid   $1.49       Everyone
3        NaN        0     19.00          0   NaN       0   Everyone 10+
4        NaN        0     10.00          0  Free       0       Everyone
…        …          …         …         …    …       …              …
5840     NaN     1317     60.00    100,000  Free       0       Everyone
5862     NaN      970     71.00    100,000  Free       0       Everyone
6141     NaN     2221      2.40    500,000  Free       0       Everyone
7035     NaN      142      3.40  1,000,000  Free       0       Everyone
7175     NaN        1      3.70  1,000,000  Free       0       Everyone

                Genres      Last_Updated         Android_Ver
0               Social     July 28, 2017          4.1 and up
1            Education    April 15, 2016          3.0 and up
2      Personalization     July 11, 2018          4.2 and up
3             Strategy     June 28, 2018  Varies with device
4             Business    August 6, 2018          4.1 and up
…                    …                 …                   …
5840        Simulation    April 22, 2018          4.1 and up
5862         Parenting      May 28, 2018        4.0.3 and up
6141  Libraries & Demo   January 8, 2017          2.3 and up
7035             Tools    March 27, 2017  Varies with device
7175             Tools  January 17, 2018          7.0 and up

[1474 rows x 12 columns]
```

6.0.1 Drop Unused Columns

Challenge: Remove the columns called `Last_Updated` and `Android_Version` from the DataFrame. We will not use these columns.

[8]:
```
df_apps.drop(['Last_Updated', 'Android_Ver'], axis = 1, inplace = True)
```

[9]:
```
df_apps
```

[9]:
```
                                                App         Category  Rating  \
0                            Ak Parti Yardım Toplama           SOCIAL     NaN
1                          Ain Arabic Kids Alif Ba ta           FAMILY     NaN
2      Popsicle Launcher for Android P 9.0 launcher  PERSONALIZATION     NaN
3                           Command & Conquer: Rivals           FAMILY     NaN
4                                        CX Network         BUSINESS     NaN
…                                                  …                …     …
10836                                Subway Surfers             GAME    4.50
10837                                Subway Surfers             GAME    4.50
10838                                Subway Surfers             GAME    4.50
10839                                Subway Surfers             GAME    4.50
10840                                Subway Surfers             GAME    4.50

        Reviews  Size_MBs       Installs  Type   Price Content_Rating  \
0             0      8.70              0  Paid  $13.99           Teen
1             0     33.00              0  Paid   $2.99       Everyone
2             0      5.50              0  Paid   $1.49       Everyone
3             0     19.00              0   NaN       0   Everyone 10+
4             0     10.00              0  Free       0       Everyone
…         …       …           …    …     …               …
10836  27723193     76.00  1,000,000,000  Free       0   Everyone 10+
10837  27724094     76.00  1,000,000,000  Free       0   Everyone 10+
10838  27725352     76.00  1,000,000,000  Free       0   Everyone 10+
10839  27725352     76.00  1,000,000,000  Free       0   Everyone 10+
10840  27711703     76.00  1,000,000,000  Free       0   Everyone 10+

                Genres
0               Social
1            Education
2      Personalization
3             Strategy
4             Business
…                  …
10836           Arcade
10837           Arcade
10838           Arcade
10839           Arcade
10840           Arcade

[10841 rows x 10 columns]
```

6.0.2 Find and Remove NaN values in Ratings

Challenge: How may rows have a NaN value (not-a-number) in the Ratings column? Create DataFrame called `df_apps_clean` that does not include these rows.

[10]:
```
df_apps.isna().sum()
```

[10]:
```
App                0
Category           0
Rating          1474
Reviews            0
Size_MBs           0
Installs           0
Type               1
Price              0
Content_Rating     0
Genres             0
dtype: int64
```

[11]:
```
print(f"There are {df_apps.Rating.isna().sum()} NaN values in the Ratings column")
```

```
There are 1474 NaN values in the Ratings column
```

[5]:
```
df_apps_clean = df_apps.dropna()
```

[6]:
```
df_apps.shape
```

[6]:
```
(10841, 12)
```

[7]:
```
df_apps_clean.shape
```

[7]:
```
(9365, 12)
```

[8]:
```
df_apps_clean
```

[8]:

	App	Category	Rating	Reviews	Size_MBs	Installs	Type	Price	Content_Rating	Genres
21	KBA-EZ Health Guide	MEDICAL	5.00	4	25.00	1	Free	0	Everyone	Medical
28	Ra Ga Ba	GAME	5.00	2	20.00	1	Paid	$1.49	Everyone	Arcade
47	Mu.F.O.	GAME	5.00	2	16.00	1	Paid	$0.99	Everyone	Arcade
82	Brick Breaker BR	GAME	5.00	7	19.00	5	Free	0	Everyone	Arcade
99	Anatomy & Physiology Vocabulary Exam Review App	MEDICAL	5.00	1	4.60	5	Free	0	Everyone	Medical
…	…	…	…	…	…	…	…	…	…	…
10836	Subway Surfers	GAME	4.50	27722264	76.00	100000000	Free	0	Everyone 10+	Arcade
10837	Subway Surfers	GAME	4.50	27722264	76.00	100000000	Free	0	Everyone 10+	Arcade
10838	Subway Surfers	GAME	4.50	27722264	76.00	100000000	Free	0	Everyone 10+	Arcade
10839	Subway Surfers	GAME	4.50	27722264	76.00	100000000	Free	0	Everyone 10+	Arcade
10840	Subway Surfers	GAME	4.50	27722264	76.00	100000000	Free	0	Everyone 10+	Arcade

```
…          …       …              …    …    …              …       …
10836  27723193     76.00  1,000,000,000  Free      0   Everyone 10+   Arcade
10837  27724094     76.00  1,000,000,000  Free      0   Everyone 10+   Arcade
10838  27725352     76.00  1,000,000,000  Free      0   Everyone 10+   Arcade
10839  27725352     76.00  1,000,000,000  Free      0   Everyone 10+   Arcade
10840  27711703     76.00  1,000,000,000  Free      0   Everyone 10+   Arcade

           Last_Updated   Android_Ver
21       August 2, 2018  4.0.3 and up
28     February 8, 2017    2.3 and up
47        March 3, 2017    2.3 and up
82        July 23, 2018    4.1 and up
99       August 2, 2018    4.0 and up
…                   …           …
10836     July 12, 2018    4.1 and up
10837     July 12, 2018    4.1 and up
10838     July 12, 2018    4.1 and up
10839     July 12, 2018    4.1 and up
10840     July 12, 2018    4.1 and up

[9365 rows x 12 columns]
```

6.0.3 Find and Remove Duplicates

Challenge: Are there any duplicates in data? Check for duplicates using the .duplicated() function. How many entries can you find for the "Instagram" app? Use .drop_duplicates() to remove any duplicates from `df_apps_clean`.

[15]:
```
df_apps_clean.duplicated()
```

[15]:
```
21       False
28       False
47       False
82       False
99       False
         …
10836    False
10837    False
10838    False
10839     True
10840    False
Length: 9365, dtype: bool
```

[17]:
```
df_apps_clean[df_apps_clean.duplicated()]
```

[17]:
```
                                        App            Category  Rating  \
946                    420 BZ Budeze Delivery             MEDICAL    5.00
1133                               MouseMingle              DATING    2.70
```

```
1196   Cardiac diagnosis (heart rate, arrhythmia)             MEDICAL    4.40
1231                                Sway Medical              MEDICAL    5.00
1247               Chat Kids - Chat Room For Kids              DATING    4.70
…                                                …                …     …
10802               Skype - free IM & video calls       COMMUNICATION    4.10
10809                                   Instagram             SOCIAL    4.50
10826                                Google Drive        PRODUCTIVITY    4.40
10832                                Google News  NEWS_AND_MAGAZINES    3.90
10839                             Subway Surfers                GAME    4.50

        Reviews  Size_MBs        Installs  Type   Price Content_Rating  \
946           2     11.00             100  Free       0     Mature 17+
1133          3      3.90             100  Free       0     Mature 17+
1196          8      6.50             100  Paid  $12.99       Everyone
1231          3     22.00             100  Free       0       Everyone
1247          6      4.90             100  Free       0     Mature 17+
…         …        …              …    …      …               …
10802  10484169      3.50  1,000,000,000  Free       0       Everyone
10809  66577313      5.30  1,000,000,000  Free       0           Teen
10826   2731211      4.00  1,000,000,000  Free       0       Everyone
10832    877635     13.00  1,000,000,000  Free       0           Teen
10839  27725352     76.00  1,000,000,000  Free       0   Everyone 10+

                Genres    Last_Updated         Android_Ver
946             Medical    June 6, 2018         4.1 and up
1133             Dating   July 17, 2018         4.4 and up
1196            Medical   July 25, 2018         3.0 and up
1231            Medical   July 25, 2018         5.0 and up
1247             Dating   July 24, 2018       4.0.3 and up
…                   …              …                  …
10802     Communication  August 3, 2018  Varies with device
10809            Social   July 31, 2018  Varies with device
10826      Productivity  August 6, 2018  Varies with device
10832  News & Magazines  August 1, 2018         4.4 and up
10839            Arcade   July 12, 2018         4.1 and up

[474 rows x 12 columns]
```

[17]:
```
duplicates_instagram = df_apps_clean[df_apps_clean['App'] == 'Instagram']
```

[18]:
```
duplicates_instagram
```

[18]:
```
             App Category  Rating   Reviews  Size_MBs        Installs  Type  \
10806  Instagram   SOCIAL     4.5  66577313       5.3  1,000,000,000  Free
10808  Instagram   SOCIAL     4.5  66577446       5.3  1,000,000,000  Free
10809  Instagram   SOCIAL     4.5  66577313       5.3  1,000,000,000  Free
10810  Instagram   SOCIAL     4.5  66509917       5.3  1,000,000,000  Free
```

```
      Price Content_Rating  Genres
10806     0           Teen  Social
10808     0           Teen  Social
10809     0           Teen  Social
10810     0           Teen  Social
```

[19]:
```
# df_apps_clean = df_apps_clean.drop_duplicates() Will not work ok because if␣
↪some other column is different they won't be cleared
```

[20]:
```
#instead do this :
df_apps_clean = df_apps_clean.drop_duplicates(subset = ['App', 'Type', 'Price'])
```

[21]:
```
duplicates_instagram = df_apps_clean[df_apps_clean['App'] == 'Instagram']
```

[22]:
```
duplicates_instagram
```

[22]:
```
             App Category  Rating   Reviews  Size_MBs       Installs  Type  \
10806  Instagram   SOCIAL    4.50  66577313      5.30  1,000,000,000  Free

      Price Content_Rating  Genres
10806     0           Teen  Social
```

[23]:
```
df_apps_clean.shape #Itan (9367,10)
```

[23]:
```
(8199, 10)
```

7 Find Highest Rated Apps

Challenge: Identify which apps are the highest rated. What problem might you encounter if you rely exclusively on ratings alone to determine the quality of an app?

[24]:
```
apps_5_stars = df_apps_clean[df_apps_clean['Rating'] == 5]
```

[25]:
```
apps_5_stars
```

[25]:
```
                                                App    Category  Rating  \
21                                KBA-EZ Health Guide     MEDICAL    5.00
28                                          Ra Ga Ba        GAME    5.00
47                                           Mu.F.O.        GAME    5.00
82                                 Brick Breaker BR        GAME    5.00
99   Anatomy & Physiology Vocabulary Exam Review App     MEDICAL    5.00
…                                                  …         …     …
2680                              Florida Wildflowers      FAMILY    5.00
2750  Superheroes, Marvel, DC, Comics, TV, Movies News      COMICS    5.00
3030           CL Keyboard - Myanmar Keyboard (No Ads)       TOOLS    5.00
3115                                       Oración CX   LIFESTYLE    5.00
```

```
4058                          Ek Bander Ne Kholi Dukan     FAMILY    5.00

      Reviews  Size_MBs Installs  Type  Price Content_Rating         Genres
21          4     25.00        1  Free      0       Everyone        Medical
28          2     20.00        1  Paid  $1.49       Everyone         Arcade
47          2     16.00        1  Paid  $0.99       Everyone         Arcade
82          7     19.00        5  Free      0       Everyone         Arcade
99          1      4.60        5  Free      0       Everyone        Medical
…       …        …       …    …     …             …              …
2680        5     69.00    1,000  Free      0       Everyone      Education
2750       34     12.00    5,000  Free      0       Everyone         Comics
3030       24      3.20    5,000  Free      0       Everyone          Tools
3115      103      3.80    5,000  Free      0       Everyone      Lifestyle
4058       10      3.00   10,000  Free      0       Everyone  Entertainment

[271 rows x 10 columns]
```

[26]:
```
apps_5_stars.sort_values('Reviews', ascending = False)  #All 5 star reviews␣
↪apps have a very small ammount of votes
```

[26]:
```
                                                App     Category  Rating  \
2095                                      Ríos de Fe    LIFESTYLE    5.00
2438  FD Calculator (EMI, SIP, RD & Loan Eligilibility)      FINANCE    5.00
3115                                      Oración CX    LIFESTYLE    5.00
2107                     Barisal University App-BU Face       FAMILY    5.00
2069                                      Master E.K       FAMILY    5.00
…                                                  …          …     …
1272                                      CG Prints  PHOTOGRAPHY    5.00
453                                 Wowkwis aq Ka'qaquj       FAMILY    5.00
349                                       DF Glue Board    PARENTING    5.00
230   NCLEX Multi-topic Nursing Exam Review-Quiz & n…      MEDICAL    5.00
411                                       EC SPORTS       SPORTS    5.00

      Reviews  Size_MBs Installs  Type Price Content_Rating  \
2095      141     15.00    1,000  Free     0       Everyone
2438      104      2.30    1,000  Free     0       Everyone
3115      103      3.80    5,000  Free     0       Everyone
2107      100     10.00    1,000  Free     0       Everyone
2069       90     19.00    1,000  Free     0       Everyone
…       …        …       …    …     …             …
1272        1      2.30      100  Free     0       Everyone
453         1     49.00       10  Free     0       Everyone
349         1     27.00       10  Free     0       Everyone
230         1      4.20       10  Free     0       Everyone
411         1      6.30       10  Free     0       Everyone

                  Genres
```

```
2095            Lifestyle
2438              Finance
3115            Lifestyle
2107            Education
2069            Education
…                     …
1272          Photography
453   Education;Education
349             Parenting
230               Medical
411                Sports

[271 rows x 10 columns]
```

8 Find 5 Largest Apps in terms of Size (MBs)

Challenge: What's the size in megabytes (MB) of the largest Android apps in the Google Play Store. Based on the data, do you think there could be limit in place or can developers make apps as large as they please?

[27]:
```
df_apps_clean.sort_values('Size_MBs', ascending = False)
```

[27]:
```
                                App            Category  Rating  Reviews  \
9942   Talking Babsy Baby: Baby Games           LIFESTYLE    4.00   140995
10687          Hungry Shark Evolution                GAME    4.50  6074334
9943            Miami crime simulator                GAME    4.00   254518
9944     Gangster Town: Vice District              FAMILY    4.30    65146
3144                     Vi Trainer  HEALTH_AND_FITNESS    3.60      124
…                                …                   …     …       …
2648      Ad Remove Plugin for App2SD        PRODUCTIVITY    4.10       66
5798                  ExDialer PRO Key       COMMUNICATION    4.50     5474
2684     My baby firework (Remove ad)              FAMILY    4.10       30
7966             Market Update Helper  LIBRARIES_AND_DEMO    4.10    20145
4696              Essential Resources  LIBRARIES_AND_DEMO    4.60      237

       Size_MBs     Installs  Type  Price Content_Rating  \
9942     100.00   10,000,000  Free      0       Everyone
10687    100.00  100,000,000  Free      0           Teen
9943     100.00   10,000,000  Free      0     Mature 17+
9944     100.00   10,000,000  Free      0     Mature 17+
3144     100.00        5,000  Free      0       Everyone
…         …           …   …    …             …
2648       0.02        1,000  Paid  $1.29       Everyone
5798       0.02      100,000  Paid  $3.99       Everyone
2684       0.01        1,000  Paid  $0.99       Everyone
7966       0.01    1,000,000  Free      0       Everyone
4696       0.01       50,000  Free      0       Everyone
```

```
                        Genres
9942   Lifestyle;Pretend Play
10687                  Arcade
9943                   Action
9944               Simulation
3144         Health & Fitness
…                          …
2648              Productivity
5798             Communication
2684             Entertainment
7966         Libraries & Demo
4696         Libraries & Demo

[8199 rows x 10 columns]
```

9 Find the 5 App with Most Reviews

Challenge: Which apps have the highest number of reviews? Are there any paid apps among the top 50?

[28]:
```
df_apps_clean.sort_values('Reviews', ascending = False)
```

[28]:
```
                                                  App            Category  \
10805                                        Facebook              SOCIAL
10785                                WhatsApp Messenger       COMMUNICATION
10806                                       Instagram              SOCIAL
10784       Messenger – Text and Video Chat for Free       COMMUNICATION
10650                                 Clash of Clans                GAME
…                                                 …                …
453                               Wowkwis aq Ka'qaquj              FAMILY
462                                           CB Fit  HEALTH_AND_FITNESS
901                   ES Billing System (Offline App)        PRODUCTIVITY
1416   Ek Kahani Aisi Bhi Season 3 - The Horror Story              FAMILY
1100                                  Chenoweth AH              MEDICAL

       Rating   Reviews  Size_MBs       Installs  Type Price Content_Rating  \
10805    4.10  78158306      5.30  1,000,000,000  Free     0           Teen
10785    4.40  69119316      3.50  1,000,000,000  Free     0       Everyone
10806    4.50  66577313      5.30  1,000,000,000  Free     0           Teen
10784    4.00  56642847      3.50  1,000,000,000  Free     0       Everyone
10650    4.60  44891723     98.00    100,000,000  Free     0   Everyone 10+
…         …         …         …              …     …    …            …
453      5.00         1     49.00             10  Free     0       Everyone
462      5.00         1      7.80             10  Free     0       Everyone
901      5.00         1      4.20            100  Free     0       Everyone
1416     3.00         1      5.80            100  Free     0           Teen
```

```
1100     5.00         1     27.00            100  Free     0       Everyone

                   Genres
10805              Social
10785       Communication
10806              Social
10784       Communication
10650            Strategy
…                     …
453    Education;Education
462       Health & Fitness
901           Productivity
1416        Entertainment
1100              Medical

[8199 rows x 10 columns]
```

10 Plotly Pie and Donut Charts - Visualise Categorical Data: Content Ratings

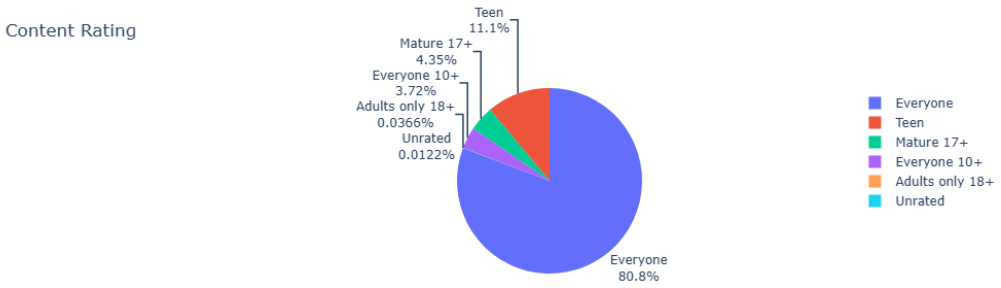
[29]:
```
ratings = df_apps_clean.Content_Rating.value_counts()
```

[30]:
```
ratings
```

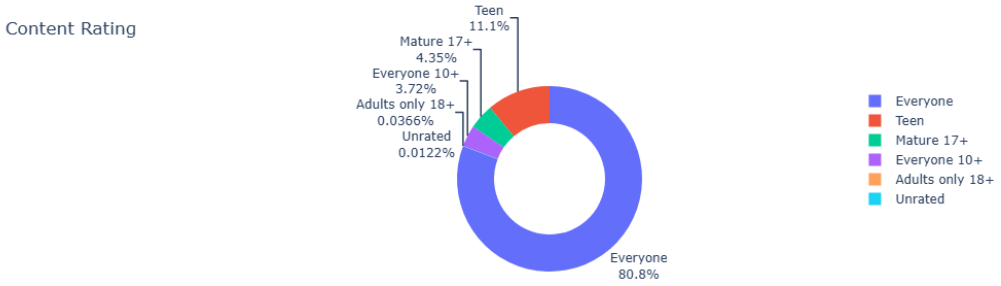
[30]:
```
Content_Rating
Everyone           6621
Teen                912
Mature 17+          357
Everyone 10+        305
Adults only 18+       3
Unrated               1
Name: count, dtype: int64
```

[31]:
```
import plotly.express as px
```

[32]:
```
piechart = px.pie(labels = ratings.index, values = ratings.values, title =␣
 ↪'Content Rating', names = ratings.index,)
piechart.update_traces(textposition = 'outside', textinfo='percent+label')
piechart.show()
```



[33]:
```
#add the parameter hole = (0 to 1) to create a donut chart
piechart = px.pie(labels = ratings.index, values = ratings.values, title =␣
↪'Content Rating', names = ratings.index, hole = 0.6)
piechart.update_traces(textposition = 'outside', textinfo='percent+label')
piechart.show()
```



11 Numeric Type Conversion: Examine the Number of Installs

Challenge: How many apps had over 1 billion (that's right - BILLION) installations? How many apps just had a single install?

Check the datatype of the Installs column.

Count the number of apps at each level of installations.

Convert the number of installations (the Installs column) to a numeric data type. Hint: this is a 2-step process. You'll have make sure you remove non-numeric characters first.

[34]:
```
df_apps_clean.Installs.describe()
```

[34]:
```
count          8199
unique           19
top       1,000,000
freq           1417
Name: Installs, dtype: object
```

[35]:
```
df_apps_clean.info() #enallaktika
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 8199 entries, 21 to 10835
Data columns (total 10 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   App             8199 non-null   object
 1   Category        8199 non-null   object
 2   Rating          8199 non-null   float64
 3   Reviews         8199 non-null   int64
 4   Size_MBs        8199 non-null   float64
 5   Installs        8199 non-null   object
 6   Type            8199 non-null   object
 7   Price           8199 non-null   object
 8   Content_Rating  8199 non-null   object
 9   Genres          8199 non-null   object
dtypes: float64(2), int64(1), object(7)
memory usage: 704.6+ KB
```

[36]:
```
df_apps_clean[['App', 'Installs']].groupby('Installs').count()    #de mas␣
↪aresoun ta komma
```

[36]:

Installs	App
1	3
1,000	698
1,000,000	1417
1,000,000,000	20
10	69
10,000	988
10,000,000	933
100	303
100,000	1096
100,000,000	189
5	9
5,000	425
5,000,000	607
50	56
50,000	457
50,000,000	202
500	199

```
500,000          504
500,000,000       24
```

[37]:
```
df_apps_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 8199 entries, 21 to 10835
Data columns (total 10 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   App             8199 non-null   object
 1   Category        8199 non-null   object
 2   Rating          8199 non-null   float64
 3   Reviews         8199 non-null   int64
 4   Size_MBs        8199 non-null   float64
 5   Installs        8199 non-null   object
 6   Type            8199 non-null   object
 7   Price           8199 non-null   object
 8   Content_Rating  8199 non-null   object
 9   Genres          8199 non-null   object
dtypes: float64(2), int64(1), object(7)
memory usage: 704.6+ KB
```

[38]:
```
df_apps_clean.Installs = df_apps_clean.Installs.astype(str).str.replace(',',"")
```

```
C:\Users\user\AppData\Local\Temp\ipykernel_8840\70812682.py:1:
SettingWithCopyWarning:


A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
```

[39]:
```
df_apps_clean.Installs = pd.to_numeric(df_apps_clean.Installs)
```

```
C:\Users\user\AppData\Local\Temp\ipykernel_8840\665362016.py:1:
SettingWithCopyWarning:


A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
```

[40]:
```
df_apps_clean[['App', 'Installs']].groupby('Installs').count()
```

[40]:

Installs	App
1	3
5	9
10	69
50	56
100	303
500	199
1000	698
5000	425
10000	988
50000	457
100000	1096
500000	504
1000000	1417
5000000	607
10000000	933
50000000	202
100000000	189
500000000	24
1000000000	20

12 Find the Most Expensive Apps, Filter out the Junk, and Calculate a (ballpark) Sales Revenue Estimate

Let's examine the Price column more closely.

Challenge: Convert the price column to numeric data. Then investigate the top 20 most expensive apps in the dataset.

Remove all apps that cost more than $250 from the `df_apps_clean` DataFrame.

Add a column called 'Revenue_Estimate' to the DataFrame. This column should hold the price of the app times the number of installs. What are the top 10 highest grossing paid apps according to this estimate? Out of the top 10 highest grossing paid apps, how many are games?

[41]:
```
df_apps_clean['Price'].describe()
```

[41]:
```
count     8199
unique      73
top          0
freq      7595
Name: Price, dtype: object
```

[42]:
```
df_apps_clean.Price = df_apps_clean.Price.astype(str).str.replace('$',"")
df_apps_clean.Price = pd.to_numeric(df_apps_clean.Price)
```

```
df_apps_clean.sort_values('Price', ascending = False).head(20)
```

```
C:\Users\user\AppData\Local\Temp\ipykernel_8840\3837063921.py:1:
SettingWithCopyWarning:


A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

C:\Users\user\AppData\Local\Temp\ipykernel_8840\3837063921.py:2:
SettingWithCopyWarning:


A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
```

[42]:

	App	Category	Rating	Reviews	Size_MBs	Installs	Type	Price	Content_Rating	Genres
3946	I'm Rich - Trump Edition	LIFESTYLE	3.60	275	7.30	10000	Paid	400.00	Everyone	Lifestyle
2461	I AM RICH PRO PLUS	FINANCE	4.00	36	41.00					
4606	I Am Rich Premium	FINANCE	4.10	1867	4.70					
3145	I am rich(premium)	FINANCE	3.50	472	0.94					
3554	I'm rich	LIFESTYLE	3.80	718	26.00					
5765	I am rich	LIFESTYLE	3.80	3547	1.80					
1946	I am rich (Most expensive app)	FINANCE	4.10	129	2.70					
2775	I Am Rich Pro	FAMILY	4.40	201	2.70					
3221	I am Rich Plus	FAMILY	4.00	856	8.70					
3114	I am Rich	FINANCE	4.30	180	3.80					
1331	most expensive app (H)	FAMILY	4.30	6	1.50					
2394	I am Rich!	FINANCE	3.80	93	22.00					
3897	I Am Rich	FAMILY	3.60	217	4.90					
2193	I am extremely Rich	LIFESTYLE	2.90	41	2.90					
3856	I am rich VIP	LIFESTYLE	3.80	411	2.60					
2281	Vargo Anesthesia Mega App	MEDICAL	4.60	92	32.00					
1407	LTC AS Legal	MEDICAL	4.00	6	1.30					
2629	I am Rich Person	LIFESTYLE	4.20	134	1.80					
2481	A Manual of Acupuncture	MEDICAL	3.50	214	68.00					
4264	Golfshot Plus: Golf GPS	SPORTS	4.10	3387	25.00					

```
2461      1000  Paid 399.99       Everyone        Finance
4606     50000  Paid 399.99       Everyone        Finance
3145      5000  Paid 399.99       Everyone        Finance
3554     10000  Paid 399.99       Everyone      Lifestyle
5765    100000  Paid 399.99       Everyone      Lifestyle
1946      1000  Paid 399.99           Teen        Finance
2775      5000  Paid 399.99       Everyone  Entertainment
3221     10000  Paid 399.99       Everyone  Entertainment
3114      5000  Paid 399.99       Everyone        Finance
1331       100  Paid 399.99       Everyone  Entertainment
2394      1000  Paid 399.99       Everyone        Finance
3897     10000  Paid 389.99       Everyone  Entertainment
2193      1000  Paid 379.99       Everyone      Lifestyle
3856     10000  Paid 299.99       Everyone      Lifestyle
2281      1000  Paid  79.99       Everyone        Medical
1407       100  Paid  39.99       Everyone        Medical
2629      1000  Paid  37.99       Everyone      Lifestyle
2481      1000  Paid  33.99       Everyone        Medical
4264     50000  Paid  29.99       Everyone         Sports
```

12.0.1 The most expensive apps sub $250

[43]:
```
df_apps_clean = df_apps_clean[df_apps_clean['Price'] < 250]
df_apps_clean.sort_values('Price', ascending=False).head(5)
```

[43]:
```
                           App   Category  Rating  Reviews  Size_MBs  \
2281  Vargo Anesthesia Mega App    MEDICAL    4.60       92     32.00
1407               LTC AS Legal    MEDICAL    4.00        6      1.30
2629          I am Rich Person  LIFESTYLE    4.20      134      1.80
2481    A Manual of Acupuncture    MEDICAL    3.50      214     68.00
2463         PTA Content Master    MEDICAL    4.20       64     41.00

      Installs  Type  Price Content_Rating     Genres
2281      1000  Paid  79.99       Everyone    Medical
1407       100  Paid  39.99       Everyone    Medical
2629      1000  Paid  37.99       Everyone  Lifestyle
2481      1000  Paid  33.99       Everyone    Medical
2463      1000  Paid  29.99       Everyone    Medical
```

12.0.2 Highest Grossing Paid Apps (ballpark estimate)

[44]:
```
df_apps_clean['Revenue Estimate'] = df_apps_clean.Installs.mul(df_apps_clean.
↪Price)
df_apps_clean.sort_values('Revenue Estimate', ascending = False)[:10]
```

[44]:
```
                            App     Category  Rating  Reviews  Size_MBs  \
9220                  Minecraft       FAMILY    4.50  2376564     19.00
```

```
8825                  Hitman Sniper         GAME    4.60   408292     29.00
7151  Grand Theft Auto: San Andreas         GAME    4.40   348962     26.00
7477            Facetune - For Free  PHOTOGRAPHY    4.40    49553     48.00
7977        Sleep as Android Unlock    LIFESTYLE    4.50    23966      0.85
6594            DraStic DS Emulator         GAME    4.60    87766     12.00
6082                  Weather Live      WEATHER    4.50    76593      4.75
7954                   Bloons TD 5       FAMILY    4.60   190086     94.00
7633        Five Nights at Freddy's         GAME    4.60   100805     50.00
6746     Card Wars - Adventure Time       FAMILY    4.30   129603     23.00

      Installs  Type  Price Content_Rating                     Genres  \
9220  10000000  Paid   6.99   Everyone 10+  Arcade;Action & Adventure
8825  10000000  Paid   0.99     Mature 17+                     Action
7151   1000000  Paid   6.99     Mature 17+                     Action
7477   1000000  Paid   5.99       Everyone                 Photography
7977   1000000  Paid   5.99       Everyone                   Lifestyle
6594   1000000  Paid   4.99       Everyone                      Action
6082    500000  Paid   5.99       Everyone                     Weather
7954   1000000  Paid   2.99       Everyone                    Strategy
7633   1000000  Paid   2.99           Teen                      Action
6746   1000000  Paid   2.99   Everyone 10+    Card;Action & Adventure

      Revenue Estimate
9220     69,900,000.00
8825      9,900,000.00
7151      6,990,000.00
7477      5,990,000.00
7977      5,990,000.00
6594      4,990,000.00
6082      2,995,000.00
7954      2,990,000.00
7633      2,990,000.00
6746      2,990,000.00
```

13 Plotly Bar Charts & Scatter Plots: Analysing App Categories

[45]:
```
df_apps_clean.Category.nunique()
```

[45]:
```
33
```

[46]:
```
df_apps_clean.Category.value_counts()
```

[46]:
```
Category
FAMILY                 1606
GAME                    910
TOOLS                   719
```

```
PRODUCTIVITY            301
PERSONALIZATION         298
LIFESTYLE               297
FINANCE                 296
MEDICAL                 292
PHOTOGRAPHY             263
BUSINESS                262
SPORTS                  260
COMMUNICATION           257
HEALTH_AND_FITNESS      243
NEWS_AND_MAGAZINES      204
SOCIAL                  203
TRAVEL_AND_LOCAL        187
SHOPPING                180
BOOKS_AND_REFERENCE     169
VIDEO_PLAYERS           148
DATING                  134
EDUCATION               118
MAPS_AND_NAVIGATION     118
ENTERTAINMENT           102
FOOD_AND_DRINK           94
AUTO_AND_VEHICLES        73
WEATHER                  72
LIBRARIES_AND_DEMO       64
HOUSE_AND_HOME           62
ART_AND_DESIGN           61
COMICS                   54
PARENTING                50
EVENTS                   45
BEAUTY                   42
Name: count, dtype: int64
```

[47]:
```
top10_category = df_apps_clean.Category.value_counts()[:10]
```

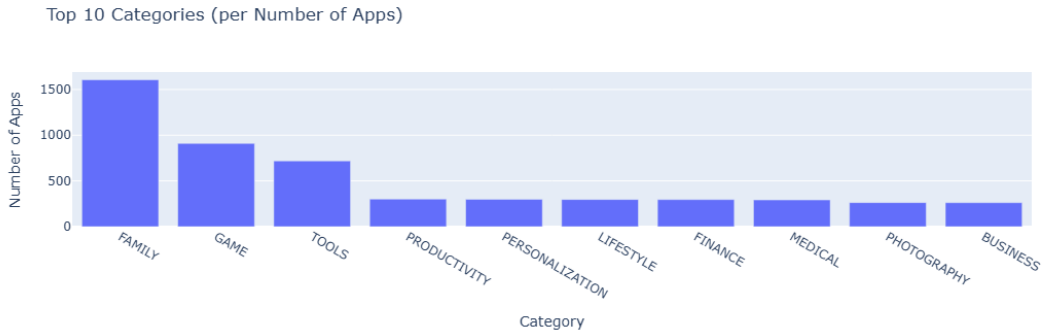
[48]:
```
top10_category
```

[48]:
```
Category
FAMILY              1606
GAME                 910
TOOLS                719
PRODUCTIVITY         301
PERSONALIZATION      298
LIFESTYLE            297
FINANCE              296
MEDICAL              292
PHOTOGRAPHY          263
BUSINESS             262
```

```
Name: count, dtype: int64
```

13.0.1 Vertical Bar Chart - Highest Competition (Number of Apps)

[49]:
```
bar = px.bar(x = top10_category.index, y = top10_category.values, title = 'Top␣
↪10 Categories (per Number of Apps)')
bar.update_layout(xaxis_title = 'Category', yaxis_title = 'Number of Apps')
bar.show()
```



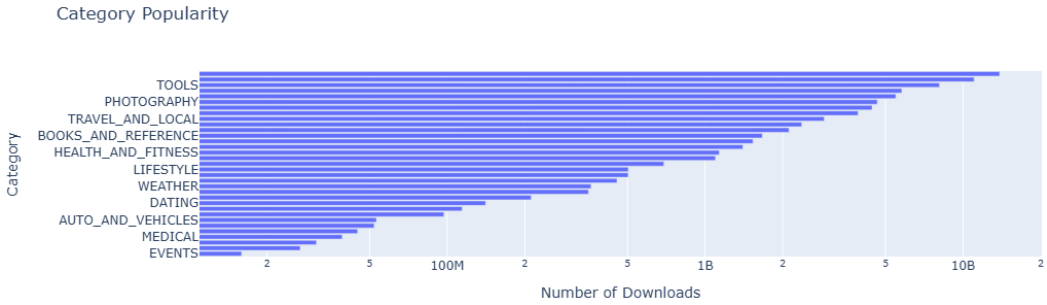
13.0.2 Horizontal Bar Chart - Most Popular Categories (Highest Downloads)

[50]:
```
category_installs = df_apps_clean.groupby('Category').agg({'Installs':pd.Series.
↪sum})
category_installs.sort_values('Installs', ascending = True, inplace = True)
category_installs.head(5)
```

[50]:

Category	Installs
EVENTS	15949410
BEAUTY	26916200
PARENTING	31116110
MEDICAL	39162676
COMICS	44931100

[74]:
```
h_bar = px.bar(x = category_installs.Installs, y = category_installs.index,␣
↪orientation='h', title = 'Category Popularity')
h_bar.update_layout(xaxis_title = 'Number of Downloads', yaxis_title =␣
↪'Category', xaxis = dict(type='log'))
h_bar.show()
```



13.0.3 Category Concentration - Downloads vs. Competition

Challenge: * First, create a DataFrame that has the number of apps in one column and the number of installs in another:

- Then use the plotly express examples from the documentation alongside the .scatter() API referenceto create scatter plot that looks like this.

Hint: Use the size, hover_name and color parameters in .scatter(). To scale the yaxis, call .update_layout() and specify that the yaxis should be on a log-scale like so: yaxis=dict(type='log')

[52]:
```
cat_number = df_apps_clean.groupby('Category').agg({'App':pd.Series.count})
```

[53]:
```
cat_number.head(5)
```

[53]:

Category	App
ART_AND_DESIGN	61
AUTO_AND_VEHICLES	73
BEAUTY	42
BOOKS_AND_REFERENCE	169
BUSINESS	262

[54]:
```
cat_merged_df = pd.merge(cat_number, category_installs, on = 'Category', how =␣
↪"inner")
```

[55]:
```
print(f'The dimensions of the DataFrame are: {cat_merged_df.shape}')
```

```
The dimensions of the DataFrame are: (33, 2)
```

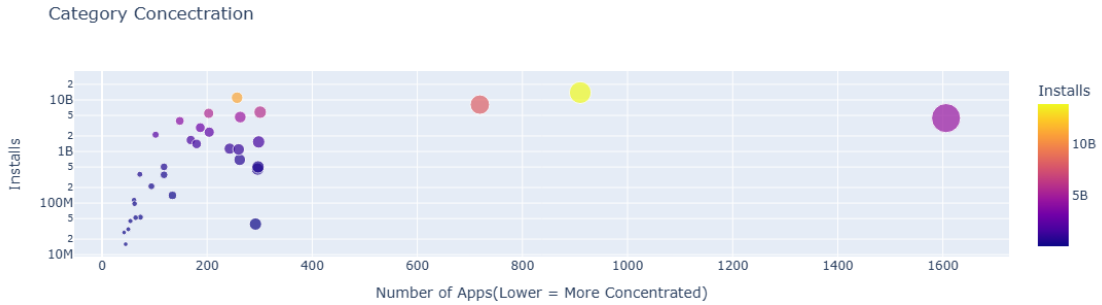
[56]:
```
cat_merged_df.sort_values('Installs', ascending=False)
```

[56]:

Category	App	Installs
GAME	910	13858762717
COMMUNICATION	257	11039241530

```
TOOLS                 719   8099724500
PRODUCTIVITY          301   5788070180
SOCIAL                203   5487841475
PHOTOGRAPHY           263   4649143130
FAMILY               1606   4437554490
VIDEO_PLAYERS         148   3916897200
TRAVEL_AND_LOCAL      187   2894859300
NEWS_AND_MAGAZINES    204   2369110650
ENTERTAINMENT         102   2113660000
BOOKS_AND_REFERENCE   169   1665791655
PERSONALIZATION       298   1532352930
SHOPPING              180   1400331540
HEALTH_AND_FITNESS    243   1134006220
SPORTS                260   1096431465
BUSINESS              262    692018120
LIFESTYLE             297    503611120
MAPS_AND_NAVIGATION   118    503267560
FINANCE               296    455249400
WEATHER                72    361096500
EDUCATION             118    352852000
FOOD_AND_DRINK         94    211677750
DATING                134    140912410
ART_AND_DESIGN         61    114233100
HOUSE_AND_HOME         62     97082000
AUTO_AND_VEHICLES      73     53129800
LIBRARIES_AND_DEMO     64     52083000
COMICS                 54     44931100
MEDICAL               292     39162676
PARENTING              50     31116110
BEAUTY                 42     26916200
EVENTS                 45     15949410
```

[57]:
```
scatter = px.scatter(cat_merged_df,
                     x = 'App',
                     y = 'Installs',
                     title = 'Category Concectration',
                     size = 'App',
                     hover_name = cat_merged_df.index,
                     color = 'Installs')
scatter.update_layout(xaxis_title = "Number of Apps(Lower = More Concentrated)",
                          yaxis_title = "Installs",
                          yaxis = dict(type='log'))
scatter.show()
```



14 Extracting Nested Data from a Column

Challenge: How many different types of genres are there? Can an app belong to more than one genre? Check what happens when you use .value_counts() on a column with nested values? See if you can work around this problem by using the .split() function and the DataFrame's .stack() method.

[114]:
```
df_apps_clean.Genres.describe()
# print(f"There are {df_apps_clean.Genres.nunique} different genres")
df_apps_clean.Genres.value_counts()
```

[114]:
```
Genres
Tools                                718
Entertainment                        467
Education                            429
Productivity                         301
Personalization                      298
                                    …
Adventure;Brain Games                  1
Travel & Local;Action & Adventure      1
Art & Design;Pretend Play              1
Music & Audio;Music & Video            1
Lifestyle;Pretend Play                 1
Name: count, Length: 114, dtype: int64
```

[116]:
```
genres = len(df_apps_clean.Genres.value_counts())
print(f"There are {genres} different genres or combinations of genres. I want␣
 ↪to find the number of the unique single genres")
```

```
There are 114 different genres or combinations of genres. I want to find the
number of the unique single genres
```

14.0.1 We need to separate the multiple genres at the colon separator.

[118]:
```
stack = df_apps_clean.Genres.str.split(';', expand = True).stack()
print(f"We now have a single column with shape: {stack.shape}")
```

```
We now have a single column with shape: (8564,)
```

[120]:
```
num_genres = stack.value_counts()
num_genres
```

[120]:
```
Tools                     719
Education                 587
Entertainment             498
Action                    304
Productivity              301
Personalization           298
Lifestyle                 298
Finance                   296
Medical                   292
Sports                    270
Photography               263
Business                  262
Communication             258
Health & Fitness          245
Casual                    216
News & Magazines          204
Social                    203
Simulation                200
Travel & Local            187
Arcade                    185
Shopping                  180
Books & Reference         171
Video Players & Editors   150
Dating                    134
Puzzle                    124
Maps & Navigation         118
Role Playing              111
Racing                    103
Action & Adventure         96
Strategy                   95
Food & Drink               94
Educational                93
Adventure                  78
Auto & Vehicles            73
Weather                    72
Pretend Play               68
Brain Games                65
Libraries & Demo           64
```

```
Art & Design                62
House & Home                62
Board                       57
Comics                      54
Parenting                   50
Card                        46
Events                      45
Beauty                      42
Casino                      37
Music & Video               31
Creativity                  31
Trivia                      28
Word                        22
Music                       21
Music & Audio                1
Name: count, dtype: int64
```

[125]:
```
print(f"Number of genres : {len(num_genres)}")
```

```
Number of genres : 53
```

15 Colour Scales in Plotly Charts - Competition in Genres

Challenge: Can you create this chart with the Series containing the genre data?

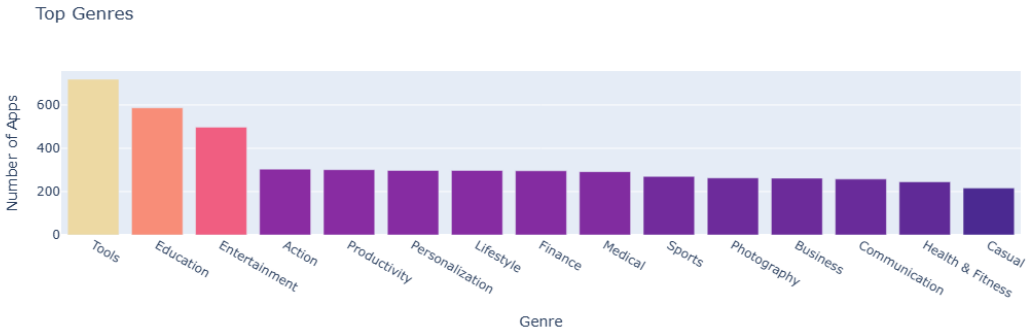
Try experimenting with the built in colour scales in Plotly. You can find a full list here.

- Find a way to set the colour scale using the color_continuous_scale parameter.
- Find a way to make the color axis disappear by using coloraxis_showscale.

[131]:
```
top_15_genres = num_genres[:15]
bar = px.bar(x = top_15_genres.index,
             y = top_15_genres.values,
             title = 'Top Genres',
             color = top_15_genres.values,
             hover_name = top_15_genres.index,
             color_continuous_scale='Agsunset')

bar.update_layout(xaxis_title = 'Genre', yaxis_title = 'Number of Apps',␣
 ↪coloraxis_showscale=False)

bar.show()
```



16 Grouped Bar Charts: Free vs. Paid Apps per Category

[132]:
```
df_apps_clean.Type.value_counts()
```

[132]:
```
Type
Free    7595
Paid     589
Name: count, dtype: int64
```

17 We see that the majority of apps are free on the Google Play Store. But perhaps some categories have more paid apps than others. Let's investigate. We can group our data first by Category and then by Type. Then we can add up the number of apps per each type. Using as_index=False we push all the data into columns rather than end up with our Categories as the index.

[144]:
```
df_free_vs_paid = df_apps_clean.groupby(['Category', 'Type'], as_index = False).
 ↪agg({'App' : pd.Series.count})
df_free_vs_paid
```

[144]:

	Category	Type	App
0	ART_AND_DESIGN	Free	58
1	ART_AND_DESIGN	Paid	3
2	AUTO_AND_VEHICLES	Free	72
3	AUTO_AND_VEHICLES	Paid	1
4	BEAUTY	Free	42
..	…	…	…
56	TRAVEL_AND_LOCAL	Paid	8
57	VIDEO_PLAYERS	Free	144

```
58      VIDEO_PLAYERS  Paid    4
59            WEATHER  Free   65
60            WEATHER  Paid    7

[61 rows x 3 columns]
```

[145]:
```
df_free_vs_paid.sort_values('App')
```

[145]:
```
              Category  Type   App
3    AUTO_AND_VEHICLES  Paid     1
24      FOOD_AND_DRINK  Paid     2
38  NEWS_AND_MAGAZINES  Paid     2
40           PARENTING  Paid     2
17       ENTERTAINMENT  Paid     2
..                 …    …   …
31           LIFESTYLE  Free   284
21             FINANCE  Free   289
53               TOOLS  Free   656
25                GAME  Free   834
19              FAMILY  Free  1456

[61 rows x 3 columns]
```

Challenge: Use the plotly express bar chart examples and the .bar() API reference to create this bar chart:

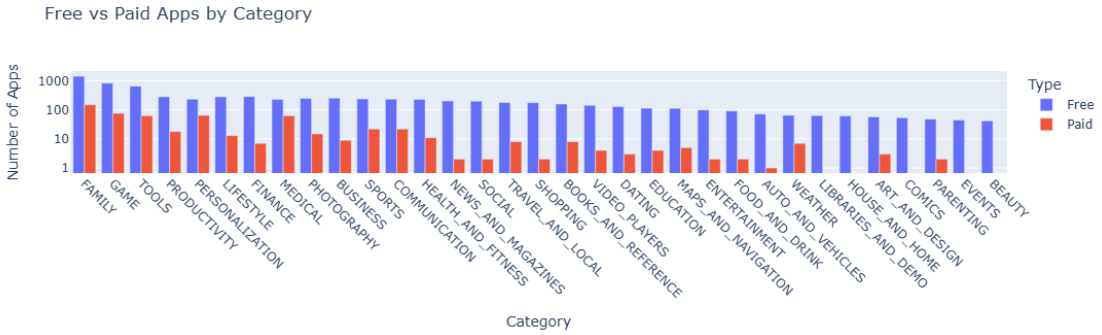
You'll want to use the `df_free_vs_paid` DataFrame that you created above that has the total number of free and paid apps per category.

See if you can figure out how to get the look above by changing the `categoryorder` to 'total descending' as outlined in the documentation here here.

[174]:
```
g_bar = px.bar(df_free_vs_paid,
               x = 'Category',
               y = 'App',
               title = 'Free vs Paid Apps by Category',
               color = 'Type',
               barmode = 'group')

g_bar.update_layout(xaxis_title = 'Category',
                    yaxis_title = 'Number of Apps',
                    xaxis_tickangle = 45,
                    xaxis = {'categoryorder' : 'total descending'},
                    yaxis = dict(type = 'log'))

g_bar.show()
```



[ ]:

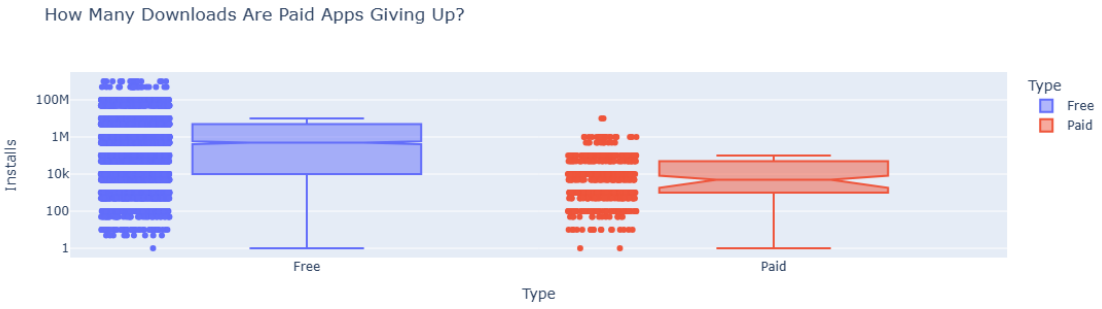
18 Plotly Box Plots: Lost Downloads for Paid Apps

Challenge: Create a box plot that shows the number of Installs for free versus paid apps. How does the median number of installations compare? Is the difference large or small?

Use the Box Plots Guide and the .box API reference to create the following chart.

[194]:
```
box = px.box(df_apps_clean,
             x = 'Type',
             y = 'Installs',
             color = 'Type',
             notched = True,
             points = 'all',
             title = 'How Many Downloads Are Paid Apps Giving Up?')
box.update_layout(yaxis = dict(type = 'log'))

box.show()
```



19 Plotly Box Plots: Revenue by App Category

Challenge: See if you can generate the chart below:

Looking at the hover text, how much does the median app earn in the Tools category? If developing an Android app costs $30,000 or thereabouts, does the average photography app recoup its development costs?

Hint: I've used 'min ascending' to sort the categories.

[216]:
```
df_paid_apps = df_apps_clean[df_apps_clean['Type'] == 'Paid']
df_paid_apps
```

[216]:

	App	Category	Rating	Reviews	Size_MBs	Installs	Type	Price	Content_Rating	Genres	Revenue Estimate
28	Ra Ga Ba	GAME	5.00	2	20.00	1	Paid	1.49	Everyone	Arcade	1.49
47	Mu.F.O.	GAME	5.00	2	16.00	1	Paid	0.99	Everyone	Arcade	0.99
233	Chess of Blades (BL/Yaoi Game) (No VA)	FAMILY	4.80	4	23.00	10	Paid	14.99	Teen	Casual	149.90
248	The DG Buddy	BUSINESS	3.70	3	11.00	10	Paid	2.49	Everyone	Business	24.90
291	AC DC Power Monitor	LIFESTYLE	5.00	1	1.20	10	Paid	3.04	Everyone	Lifestyle	30.40
…	…	…	…	…	…	…	…	…	…	…	…
7957	League of Stickman 2018- Ninja Arena PVP(Dream…	GAME	4.40	32496	99.00	1000000	Paid	0.99	Teen	Action	990,000.00
7977	Sleep as Android Unlock	LIFESTYLE	4.50	23966	0.85	1000000	Paid	5.99	Everyone	Lifestyle	5,990,000.00
7988	Where's My Water?	FAMILY	4.70	188740	69.00	1000000	Paid	1.99	Everyone	Puzzle;Brain Games	1,990,000.00
8825	Hitman Sniper	GAME	4.60	408292	29.00	10000000	Paid	0.99	Mature 17+		
9220	Minecraft	FAMILY	4.50	2376564	19.00	10000000	Paid	6.99	Everyone 10+		

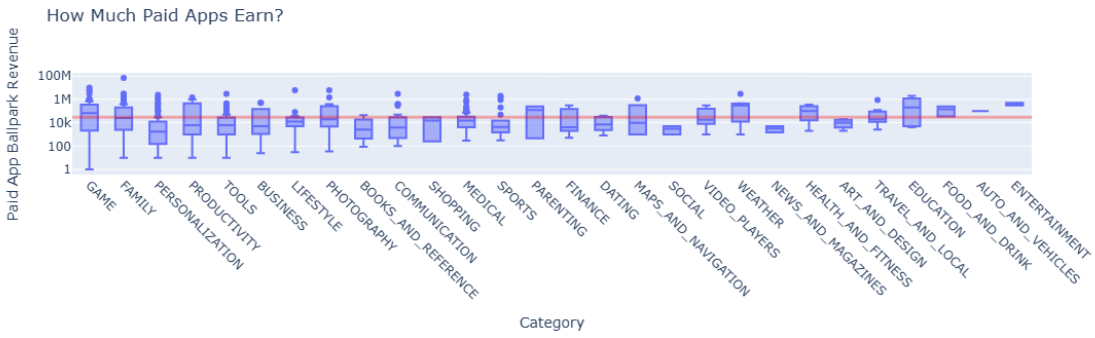
```
8825                      Action      9,900,000.00
9220  Arcade;Action & Adventure     69,900,000.00

[589 rows x 11 columns]
```

[229]:
```
box = px.box(df_paid_apps,
             x = 'Category',
             y = 'Revenue Estimate',
             title = 'How Much Paid Apps Earn?')

box.update_layout(xaxis_title = 'Category',
                  yaxis_title = 'Paid App Ballpark Revenue',
                  xaxis_tickangle = 45,
                  xaxis = {'categoryorder' : 'min ascending'},
                  yaxis = dict(type = 'log'))
box.add_hline(y=30000, line_color='red', line_width=3, opacity = 0.3)

box.show()
```



20 How Much Can You Charge? Examine Paid App Pricing Strategies by Category

Challenge: What is the median price price for a paid app? Then compare pricing by category by creating another box plot. But this time examine the prices (instead of the revenue estimates) of the paid apps. I recommend using `{categoryorder':'max descending'}` to sort the categories.

[222]:
```
df_paid_apps.Price.median()
```

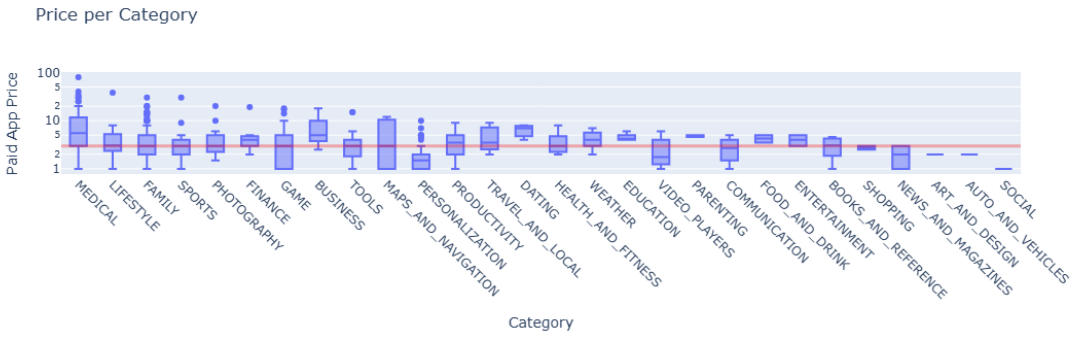
[222]: 2.99

[230]:
```
box = px.box(df_paid_apps,
             x='Category',
             y="Price",
```

```
            title='Price per Category')

box.update_layout(xaxis_title='Category',
                  yaxis_title='Paid App Price',
                  xaxis={'categoryorder':'max descending'},
                  xaxis_tickangle = 45,
                  yaxis=dict(type='log'))
box.add_hline(y=2.99 , line_color='red', line_width=3, opacity = 0.3)

box.show()
```



[ ]: